

MODULE 01

INTRODUCTION TO OPERATION RESEARCH

Introduction: Evolution of OR, Definitions of OR, Scope of OR, Applications of OR, Phases in OR, Characteristics and limitations of OR, models used in OR, Quantitative approach to decision making models (Theory Only)

Definitions:

Operations Research can be defined simply as combination of two words operation and research where operation means some action applied in any area of interest and research imply some organized process of getting and analysing information about the problem environment. However, many scientists or experts has been defined O.R. in various ways but the opinions about the definitions of it have been changed according to the growth of the subject. So before defining O.R. it is important to see few definitions of it.

O.R. is a scientific method of providing executive departments with a quantitative basis for decisions regarding the operations under their control. -Morse and Kimbal (1946)

O.R. is a scientific method of providing executive with an analytical and objective basis for decisions. P.M.S. Blackett (1948)

O.R. is the application of scientific methods, techniques and tools to problems involving the operations of system so as to provide these in control of the operations with optimum solutions to the problem. – Churchman, Acoff, Arnoff (1957)

Key Characteristics:

1. Interdisciplinary:

Combines multiple disciplines: OR combines concepts from mathematics, statistics, computer science, engineering, and economics. Integrates different fields: OR integrates different fields, such as management science, decision science, and systems engineering.

2. Analytical:

Uses mathematical models: OR uses mathematical models to analyze and solve complex problems. Employs statistical techniques: OR employs statistical techniques, such as regression

analysis and hypothesis testing. Utilizes optimization techniques: OR utilizes optimization techniques, such as linear programming and dynamic programming.

3. Problem-solving:

Focuses on problem-solving: OR focuses on solving complex problems, breaking them down into manageable parts. Uses a systematic approach: OR uses a systematic approach, involving problem definition, data collection, model building, and solution implementation. Employs creative thinking: OR employs creative thinking, using techniques such as brainstorming and mind mapping.

4. Decision-making:

Supports informed decision-making: OR provides insights and recommendations to support informed decisionmaking. Uses decision analysis: OR uses decision analysis, involving techniques such as decision trees and sensitivity analysis. Considers multiple criteria: OR considers multiple criteria, such as cost, time, and quality

5. Quantitative

Relies on quantitative data: OR relies heavily on quantitative data, using techniques such as data mining and statistical analysis. Uses mathematical models: OR uses mathematical models, such as linear programming and simulation models. Employs optimization techniques: OR employs optimization techniques, such as linear programming and dynamic programming.

6. Systematic

Follows a structured approach: OR follows a structured approach, involving problem definition, data collection, model building, and solution implementation. Uses a phased approach: OR uses a phased approach, involving phases such as problem definition, analysis, and implementation. Employs checklists and templates: OR employs checklists and templates, such as decision trees and fishbone diagrams.

7. Objective

Aims to optimize performance: OR aims to optimize performance, whether it's maximizing profit, minimizing cost, or improving quality. Uses objective criteria: OR uses objective criteria, such as cost, time, and quality, to evaluate alternatives. Avoids subjective bias: OR avoids subjective bias, using techniques such as sensitivity analysis and scenario planning.

SCOPE OF OPERATIONS RESEARCH

Scope of O.R. is very wide in today's world as it provides better solution to various decision-making problems with great speed and efficiency. Areas where methods/models developed in Operations Research can be applied are given here under:

- 1. In Agriculture:** With the explosion of population and consequent shortage of food, every country is facing the problem of optimum allocation of land to various crops in accordance with the climatic conditions, optimum distribution of water from different resources. Problems of agriculture production under various restrictions can be solved by applications of Operations Research techniques.
- 2. In Defence Operations:** Since Second World War operation research have been used for Defence operations with the aim of obtaining maximum gains with minimum efforts.
- 3. In Finance:** In these modern times, government of every country or every organisation wants to introduce such type of planning/policies regarding their finance and accounting which optimize capital investment, determine optimal replacement strategies, apply cash flow analysis for long range capital investments, formulate credit policies, credit risk. Techniques developed in O.R. can be applied for attaining above said things.
- 4. In Marketing:** A Marketing Administrator has to face many problems like production selection, formulation of competitive strategies, distribution strategies, selection of advertising media with respect to cost and time, finding the optimal number of salesmen, finding optimum time to launch a product. All such problems can be overcome using Operations Research Techniques.
- 5. In Personnel Management:** Every organization wants to make selection of personnel on minimum salary. It needs to find the best combination of workers in different categories with respect to costs, skills, age and nature of jobs. It also needs to frame recruitment policies, assign jobs to machines or workers.
- 6. In LIC:** Operations Research Techniques can be fruitfully applied in LIC offices as it enables the policy makers to decide the premium rates for various modes of policies.
- 7. In Research and Development:** In determination of the areas of concentration of research and development. It also helps in project selection.
8. O.R. helps in solving many other problems faced by public as well as private sectors such as the ones in economic and social planning, management of natural resources, energy, housing pollution control, waiting lines and administrative problems, insurance policies and many more.

ADVANTAGES OF OPERATIONS RESEARCH

Following are certain advantages of Operations Research (OR):

- **Operations Research helps decision –maker to take better and quicker decisions.** It helps decision –maker to evaluate the risk and results of all the alternative decisions. So, it improves the quality of decisions and makes the decisions more effective.
- **Operation Research helps, in preparing future managers as it provides in-depth knowledge about a particular action.** Operations Research develop models, which provides logical and systematic approach for understanding, Solving and controlling a problem.
- **Operations research reduces the chances of failure as it provides many alternatives for one problem,** which helps the management to choose the best decision. Even managers can evaluate the risks associated with each solution and can decide whether they want to go with the solution or not.
- **It helps users in optimum use of resources.** For example, linear programming techniques in Operations Research suggest most effective methods and efficient ways of optimality.
- **It helps in finding the limitations and scope of an activity.**
- Using this information, he can measure the performance of employees and can compare it with the standard performance. It modifies mathematical solutions before these are applied. Managers may accept or modify the mathematical solutions obtained using Operations Research techniques.
- It helps suggest alternative solutions for the same optimum profit if the management wants so.

LIMITATIONS OF OPERATIONS RESEARCH

Operations Research has number of applications; similarly it also has certain limitations. These limitations are mostly related to the model building and money and time factors problems involved in its application. Some of them are as given below:

➤ i)Distance between O.R. specialist and Manager

Operations Researchers job needs a mathematician or statistician, who might not be aware of the business problems. Similarly, a manager is unable to understand the complex nature of Operations Research. Thus there is a big gap between the two personnel.

➤ ii)Magnitude of Calculations

The aim of the O.R. is to find out optimal solution taking into consideration all the factors. In this modern world these factors are enormous and expressing them in quantitative model and establishing relationships among these require voluminous calculations, which can be handled only by machines.

➤ iii)Money and Time Costs

The basic data are subjected to frequent changes, incorporating these changes into the operations research models is very expensive. However, a fairly good solution at present may be more desirable than a perfect operations research solution available in future or after some time.

➤ iv)Non-quantifiable Factors

When all the factors related to a problem can be quantifiable only then operations research provides solution otherwise not. The non-quantifiable factors are not incorporated in O.R.models. Importantly O.R. models do not take into account emotional factors or qualitative factors.

➤ v)Implementation

Once the decision has been taken it should be implemented. The implementation of decisions is a delicate task. This task must take into account the complexities of human relations and behavior and in some times only the psychological factors.

Stages of Development of Operations Research

The stages of development of O.R. are also known as phases and process of O.R, which has six important steps. These six steps are arranged in the following order:

Step I: Observe the problem environment

Step II: Analyze and define the problem

Step III: Develop a model Step

IV: Select appropriate data input Step

V: Provide a solution and test its reasonableness Step

VI: Implement the solution

Step I: Observe the problem environment

The first step in the process of O.R. development is the problem environment observation. This step includes different activities; they are conferences, site visit, research, observations etc. These activities provide sufficient information to the O.R. specialists to formulate the problem.

Step II: Analyze and define the problem

This step is analyzing and defining the problem. In this step in addition to the problem definition the objectives, uses and limitations of O.R. study of the problem also defined. The outputs of this step are clear grasp of need for a solution and its nature understanding.

Step III: Develop a model

This step develops a model; a model is a representation of some abstract or real situation. The models are basically mathematical models, which describes systems, processes in the form of equations, formula/relationships. The different activities in this step are variables definition, formulating equations etc. The model is tested in the field under different environmental constraints and modified in order to work. Some times the model is modified to satisfy the management with the results.

Step IV: Select appropriate data input

A model works appropriately when there is appropriate data input. Hence, selecting appropriate input data is important step in the O.R. development stage or process. The activities in this step include internal/external data analysis, fact analysis, and collection of opinions and use of computer data banks. The objective of this step is to provide sufficient data input to operate and test the model developed in Step_III.

Step V: Provide a solution and test its reasonableness

This step is to get a solution with the help of model and input data. This solution is not implemented immediately, instead the solution is used to test the model and to find there is any limitations. Suppose if the solution is not reasonable or the behaviour of the model is not proper, the model is updated and modified at this stage. The output of this stage is the solution(s) that supports the current organizational objectives.

Step VI: Implement the solution

At this step the solution obtained from the previous step is implemented. The implementation of the solution involves mo many behavioural issues. Therefore, before implementation the implementation authority has to resolve the issues. A properly implemented solution results in quality of work and gains the support from the management.

O.R. Tools and Techniques

Linear Programming:

This is a constrained optimization technique, which optimize some criterion within some constraints. In Linear programming the objective function (profit, loss or return on investment) and constraints are linear. There are different methods available to solve linear programming.

Game Theory:

This is used for making decisions under conflicting situations where there are one or more players/opponents. In this the motive of the players are dichotomized. The success of one player tends to be at the cost of other players and hence they are in conflict.

Decision Theory:

Decision theory is concerned with making decisions under conditions of complete certainty about the future outcomes and under conditions such that we can make some probability about what will happen in future.

Inventory Models:

Inventory model make a decisions that minimize total inventory cost. This model successfully reduces the total cost of purchasing, carrying, and out of stock inventory.

Simulation:

Simulation is a procedure that studies a problem by creating a model of the process involved in the problem and then through a series of organized trials and error solutions attempt to determine the best solution. Some times this is a difficult/time consuming procedure. Simulation is used when actual experimentation is not feasible or solution of model is not possible.

Non-linear Programming:

This is used when the objective function and the constraints are not linear in nature. Linear relationships may be applied to approximate non-linear constraints but limited to some range, because approximation becomes poorer as the range is extended. Thus, the non-linear programming is used to determine the approximation in which a solution lies and then the solution is obtained using linear methods.

Dynamic Programming:

Dynamic programming is a method of analyzing multistage decision processes. In this each elementary decision depends on those preceding decisions and as well as external factors.

Integer Programming:

If one or more variables of the problem take integral values only then dynamic programming method is used. For example number of motor in an organization, number of passenger in an aircraft, number of generators in a power generating plant, etc.

Markov Process:

Markov process permits to predict changes over time information about the behavior of a system is known. This is used in decision making in situations where the various states are defined. The probability from one state to another state is known and depends on the current state and is independent of how we have arrived at that particular state.

Network Scheduling:

This technique is used extensively to plan, schedule, and monitor large projects (for example computer system installation, R & D design, construction, maintenance, etc.). The aim of this technique is minimize trouble spots (such as delays, interruption, production bottlenecks, etc.) by identifying the critical factors. The different activities and their relationships of the entire project are represented diagrammatically with the help of networks and arrows, which is used for identifying critical activities and path. There are two main types of technique in network scheduling, they are:

Program Evaluation and Review Technique (PERT) – is used when activities time is not known accurately/ only probabilistic estimate of time is available.

Critical Path Method (CPM) – is used when activities time is known accurately

Applications of Operations Research

Today, almost all fields of business and government utilizing the benefits of Operations Research. There are voluminous of applications of Operations Research. Although it is not feasible to cover all applications of O.R. in brief. The following are the abbreviated set of typical operations research applications to show how widely these techniques are used today:

- **Accounting:** Assigning audit teams effectively Credit policy analysis Cash flow planning Developing standard costs Establishing costs for byproducts Planning of delinquent account strategy
- **Construction:** Project scheduling, monitoring and control Determination of proper work force Deployment of work force Allocation of resources to projects

- Facilities Planning: Factory location and size decision Estimation of number of facilities required Hospital planning International logistic system design Transportation loading and unloading Warehouse location decision
- Finance: Building cash management models Allocating capital among various alternatives Building financial planning models Investment analysis Portfolio analysis Dividend policy making
- Manufacturing: Inventory control, Marketing balance projection, Production scheduling, Production smoothing
- Marketing: Advertising, budget allocation, Product introduction timing, Selection of Product mix, Deciding most effective packaging alternative

QUANTITATIVE APPROACH TO DECISION-MAKING OR/MS

Managerial decision-making is a process by which the management, when faced with a problem, chooses a specific course of action from a set of possible options. In making a decision, a business manager attempts to choose the most effective course of action in the given circumstances in attaining the goals of the organisation. The various types of decision-making situations that a manager might encounter can be listed as follows:

1. Decisions under certainty, where all facts are known fully and for sure, or under uncertainty where the event that would actually occur is not known but probabilities can be assigned to various possible occurrences.
2. Decisions for one time period only, called static decisions, or a sequence of interrelated decisions made either simultaneously or over several time periods, called dynamic decisions.
3. Decisions where the opponent is nature (digging an oil well, for example) or a rational opponent (for instance, setting the advertising strategy where the actions of competitors have to be considered)

These classes of decision-making situations are not mutually exclusive and a given situation would exhibit characteristics from each class. Stocking of an item for sale in a certain trade fair, for instance, illustrates static decision-making situation where uncertainty exists and nature is the opponent.

The elements of any decision are:

- a) A decision-maker, who could be an individual, group, organisation, or society;
- b) A set of possible actions that may be taken to solve the decision problems;

- c) a set of possible states that might occur;
- d) a set of consequences (pay-off) associated with various combinations of courses of actions and the states that may occur; and
- e) the relationship between pay off and the values of the decision-maker

In an actual decision-making situation, the definition and identification of alternatives, the states and the consequences, are most difficult, albeit not the most crucial aspects of the decisions problem to real life, some decision-making situations are simple while others are not. Complexities in decision situations arise due to several factors. These include the complicated manner of interaction of the economic, political, technological, environmental and competitive forces in society, the limited resources of an organisation, the values, risk attitudes and knowledge of the decision-makers and the like. For example, a company's decision to introduce a new product will be influenced by such considerations as market conditions, labor rates and availability, and investment requirements and availability of funds. The decision will be of multidimensional response, including the production methodology, cost and quality of the product, price, package design, and marketing and advertising strategy. The result of the decision would conceivably affect every segment of the organisation. The essential idea of the quantitative approach to decision-making is that if the factors that influence the decisions can be identified and quantified, it becomes easier to resolve the complexity of the tools of quantitative analysis. In fact, a large number of business problems have been given a quantitative representation with varying degrees of success and it has led to a general approach which is variedly designated as operations research (or operational research, management science, systems analysis, decision analysis, decision science, and so on. Quantitative analysis is now extended to several areas of business operations and represents probably the most effective approach to handling of some types of decision problems.

A significant benefit of attaining some degree of proficiency with quantitative methods is exhibited in the way problems are perceived and formulated. A problem has to be well defined before it can be formulated into a well-structured framework for solution. This requires an orderly and organised way of thinking.

Two observations may be made here. First, it should be understood clearly that, although quantitative analysis represents a scientific approach to decision-making, a decision by itself does not become a good and right decision for adoption merely because it is made within an orderly and mathematically precise framework. Quantification at best is an aid to business judgement and not its substitute. A certain degree

of constructive skepticism is as desirable in considering a quantitative analysis of business decisions as it is in any other process of decisionmaking. Further, some allowances should be made for qualitative factors involving morale, motivation, leadership, and so on, which cannot be ignored. But they should not be allowed to dominate to such an extent that the quantitative analysis seems to be an interesting, but worthless, academic exercise. In fact, the manager should seek some balance between quantitative and qualitative factors.

Secondly, it may be noted that various names for quantitative analysis operations research, management science, and so on-connote more or less the same general approach. We shall not attempt to discuss the differences among various labels, as it is prone to create more heat than light, but only state that the basic reason for so many titles is that the field is relatively new, and there is no consensus regarding which field of knowledge it includes. Here, we use the terms management science, operations research, and quantitative analysis interchangeably. We shall now briefly discuss operations research its historical development, nature and characteristics, and methodology.

MODULE -2

LINEAR PROGRAMMING

Linear Programming Problem (LPP), Generalized LPP- Formulation of LPP, Guidelines for formulation of linear programming model, Assumption, Advantages, Limitations, Linear Programming problem (LPP), optimal and feasible Solutions by graphical method (minimization and maximization), Simplex method (Theory and Problems)

INTRODUCTION TO LINEAR PROGRAMMING

Linear Programming is a problem solving approach that has been developed to help managers to make decisions.

Linear Programming is a mathematical technique for determining the optimum allocation of resources and obtaining a particular objective when there are alternative uses of the resources, money, manpower, material, machine and other facilities.

Linear programming deals with the optimization (maximization or minimization) of a function of variables known as objective functions. It is subject to a set of linear equalities and/or inequalities known as constraints. Linear programming is a mathematical technique which involves the allocation of limited resources in an optimal manner on the basis of a given criterion of optimality.

Linear Programming is a special and versatile technique which can be applied to a variety of management problems viz. Advertising, Distribution, Investment, Production, Refinery Operations, and Transportation analysis. The linear programming is useful not only in industry and business but also in non-profit sectors such as Education, Government, Hospital, and Libraries. The linear programming method is applicable in problems characterized by the presence of decision variables. The objective function and the constraints can be expressed as linear functions of the decision variables.

Characteristics:

All linear programming problems must have following five characteristics:

- **Objective function:** There must be clearly defined objective which can be stated in quantitative way. In business problems the objective is generally profit maximization or cost minimization.
- **Constraints:** All constraints (limitations) regarding resources should be fully spelt out in mathematical form.
- **Non-negativity:** The value of variables must be zero or positive and not negative. For example, in the case of production, the manager can decide about any particular product number in positive or minimum zero, not the negative.
- **Linearity:** The relationships between variables must be linear. Linear means proportional relationship between two _or more variable, i.e., the degree of variables should be maximum one.
- **Finiteness:** The number of inputs and outputs need to be finite. In the case of infinite factors, to compute feasible solution is not possible.

LINEAR PROGRAMMING ASSUMPTIONS:

1. **Proportionality:** The basic assumption underlying the linear programming is that any change in the constraint inequalities will have the proportional change in the objective function. This means, if product contributes Rs 20 towards the profit, then the total contribution would be equal to $20x_1$, where x_1 is the number of units of the product. For example, if there are 5 units of the product, then the contribution would be Rs 100 and in the case of 10 units, it would be Rs 200. Thus, if the output (sales) is doubled, the profit would also be doubled.
2. **Additivity:** The assumption of additivity asserts that the total profit of the objective function is determined by the sum of profit contributed by each product separately. Similarly, the total amount of resources used is determined by the sum of resources used by each product separately. This implies, there is no interaction between the decision variables.
3. **Continuity:** Another assumption of linear programming is that the decision variables are continuous. This means a combination of outputs can be used with the fractional values along with the integer values. For example, If $52\frac{2}{3}$ units of product A and $101\frac{1}{3}$ units of product B to be produced in a week. In this case, the fractional amount of production will be taken as a work-in-progress and the remaining production part is taken in the following week. Therefore, a production of 17 units of product A and 31 units of product B over a three-week period implies $52\frac{2}{3}$ units of product A and $101\frac{1}{3}$ units of product B per week.
4. **Certainty:** Another underlying assumption of linear programming is a certainty, i.e. the parameter of objective function coefficients and the coefficients of constraint inequalities is known with certainty. Such as profit per unit of product, availability of material and labor per unit, requirement of material and labor per unit are known and is given in the linear programming problem.
5. **Finite Choices:** This assumption implies that the decision maker has certain choices, and the decision variables assume non-negative values. The non-negative assumption is true in the sense, the output in the production problem cannot be negative. Thus, this assumption is considered feasible.

MANAGERIAL USES AND APPLICATIONS:

LP technique is applied to a wide variety of problems listed below:

- Optimizing the product mix when the production line works under certain specification;
- Securing least cost combination of inputs;

- Selecting the location of Plant;
- Deciding the transportation route;
- Utilizing the storage and distribution centers;
- Proper production scheduling and inventory control;
- Solving the blending problems; (h) Minimizing the raw materials waste;
- Assigning job to specialized personnel.

LINEAR PROGRAM STRUCTURE

Linear Programming models consist of an objective function and the constraints on that function. A linear programming model takes the following form:

Objective function:

$$Z = a_1X_1 + a_2X_2 + a_3X_3 + \dots + a_nX_n$$

Constraints:

$$b_{11}X_1 + b_{12}X_2 + b_{13}X_3 + \dots + b_{1n}X_n < c_1$$

$$b_{21}X_1 + b_{22}X_2 + b_{23}X_3 + \dots + b_{2n}X_n < c_2$$

....

$$b_{m1}X_1 + b_{m2}X_2 + b_{m3}X_3 + \dots + b_{mn}X_n < c_m$$

In this system of linear equations, Z is the objective function value that is being optimized, X_i are the decision variables whose optimal values are to be found, and a_i , b_{ij} and c_i are constants derived from the specifics of the problem.

LINEAR PROGRAMMING FORMULATION:

The procedure for mathematical formulation of an LP problem consists of the following steps:

Step 1 Write down the decision variables of the problem.

Step 2 Formulate the objective function to be optimized (Maximized or Minimized) as a linear function of the decision variables.

Step 3 Formulate the other conditions of the problem such as resource limitation, market constraints and interrelations between variables as linear in equations or equations in terms of the decision variables.

Step 4 Add the non-negativity constraint from the considerations so that the negative values of the decision variables do not have any valid physical interpretation.

GRAPHICAL METHOD

A simple linear programming problem with two decision variables can be easily solved by the graphical method.

Procedure for Solving LPP by Graphical Method

The steps involved in the graphical method are as follows.

- | | |
|--------|---|
| Step 1 | Consider each inequality constraint as an equation. |
| Step 2 | Plot each equation on the graph as each will geometrically represent a straight line. |
| Step 3 | Mark the region. If the inequality constraint corresponding to that line is O then the region below the line lying in the first quadrant (due to non-negativity of variables) is shaded. For the inequality constraint P sign, the region above the line in the first quadrant is shaded. The points lying in common region will satisfy all the constraints simultaneously. The common region, thus obtained, is called the feasible region. |
| Step 4 | Assign an arbitrary value, say zero, for the objective function. |
| Step 5 | Draw a straight line to represent the objective function with the arbitrary value (i.e., a straight line through the origin). |
| Step 6 | Stretch the objective function line till the extreme points of the feasible region. In the maximization case, this line will stop farthest from the origin, passing through at least one corner of the feasible region. In the minimization case, this line will stop nearest to the origin and passes through at least one corner of the feasible region. |
| Step 7 | Find the coordinates of the extreme points selected in step 6 and find the maximum or minimum value of Z . |

LIMITATIONS OF LPP:

1. It is not easy to define a specific objective function.
2. Even if a specific objective function is laid down, it may not be so easy to find out various technological, financial and other constraints which may be operative in pursuing the given objective.
3. Given a specific objective and a set of constraints, it is possible that the constraints may not be directly expressible as linear inequalities.
4. Even if the above problems are surmounted, a major problem is one of estimating relevant values of the various constant coefficients that enter into a linear programming mode, i.e., prices, etc.
5. This technique is based on the assumption of linear relations between inputs and outputs. This means that inputs and outputs can be added, multiplied and divided. But the relations between inputs and outputs are not always linear. In real life, most of the relations are non-linear.
6. This technique assumes perfect competition in product and factor markets. But perfect competition is not a reality.
7. The LP technique is based on the assumption of constant returns. In reality, there are either diminishing or increasing returns which a firm experiences in production.

8. It is a highly mathematical and complicated technique. The solution of a problem with linear programming requires the maximization or minimization of a clearly specified variable. The solution of a linear programming problem is also arrived at with such complicated method as the simplex method which involves a large number of mathematical calculations.

9. Mostly, linear programming models present trial-and-error solutions and it is difficult to find out really optimal solutions to the various economic problems.

MERITS OF LPP

1. Helps management to make efficient use of resources.
2. Provides quality in decision making.
3. Excellent tools for adjusting to meet changing demands.
4. Fast determination of the solution if a computer is used.
5. Provides a natural sensitivity analysis.
6. Finds solution to problems with a very large or infinite number of possible solution.

DEMERITS OF LPP

1. Existence of non-linear equation: The primary requirements of Linear Programming is the objective function and constraint function should be linear. Practically linear relationship do not exist in all cases.
2. Interaction between variables: LP fails in a situation where non-linearity in the equation emerge due to joint interaction between some of the activities like total effectiveness.
3. Fractional Value: In LPP fractional values are permitted for the decision variable.
4. Knowledge of Coefficients of the equation: It may not be possible to state all coefficients in the objective function and constraints with certainty.

Feasible Solution: Any set of non-negative allocations ($X_{ij} > 0$) which satisfies the row and column sum (rim requirement) is called a feasible solution.

Basic Feasible Solution: A feasible solution is called a basic feasible solution if the number of nonnegative allocations is equal to $m+n-1$, where m is the number of rows and n the number of columns in a transportation table.

ILLUSTRATIONS

1. A factory manufactures two products A and B. To manufacture one unit of A, 1.5 machine hours and 2.5 labour hours are required. To manufacture product B, 2.5 machine hours and 1.5 labour hours are required. In a month, 300 machine hours and 240 labour hours are available. Profit per unit for A is Rs. 50 and for B is Rs. 40. Formulate as LPP.
2. An investor is considering investing in two securities 'A' and 'B'. The risk and return associated with these securities is different. Security 'A' gives a return of 9% and has a risk factor of 5 on a scale of zero to 10. Security 'B' gives return of 15% but has risk factor of 8. Total amount to be invested is Rs. 5, 00, 000/- Total minimum returns on the investment should be 12%. Maximum combined risk should not be more than 6. Formulate as LPP.
3. An advertising agency is planning to launch an ad campaign. Media under consideration are T.V., Radio & Newspaper. Each medium has different reach potential and different cost. Minimum 10, 000, 000 households are to be reached through T.V. Expenditure on newspapers should not be more than Rs. 10, 00, 000. Total advertising budget is Rs. 20 million. Following data is available:

Medium	Cost per Unit (Rs.)	Reach per unit (No. of households)
Television	2, 00, 000	20, 00, 000
Radio	80, 000	10, 00, 000
Newspaper	40, 000	2, 00, 000

4. Vitamins B1 and B2 are found in two foods F1 and F2. 1 unit of F1 contains 3 units of B1 and 4 units of B2. 1 unit of F2 contains 5 units of B1 and 3 units of B2 respectively. Minimum daily prescribed consumption of B1 & B2 is 50 and 60 units respectively. Cost per unit of F1 & F2 is Rs. 6 & Rs. 3 respectively. Formulate as LPP.
5. A farmer has 200 acres of land. He produces three products X, Y & Z. Average yield per acre for X, Y & Z is 4000, 6000 and 2000 kg. Selling price of X, Y & Z is Rs. 2, 1.5 & 4 per kg respectively. Each product needs fertilizers. Cost of fertilizer is Rs. 1 per kg. Per acre need for fertilizer for X, Y & Z is 200, 200 & 100 kg respectively. Labour requirements for X, Y & Z is 10,

12 & 10 man hours per acre. Cost of labour is Rs. 40 per man hour. Maximum availability of labour is 20,000 man hours. Formulate as LPP to maximise profit.

Problems on solution to LPP using graphical method:

A company manufactures two products A and B. Both products are processed on two machines M_1 & M_2 .

	M_1	M_2
A	6 Hrs/Unit	2 Hrs/Unit
B	4 Hrs/Unit	4 Hrs/Unit
Availability	7200 Hrs/month	4000 Hrs/month

Profit per unit for A is Rs. 100 and for B is Rs. 80. Find out the monthly production of A and B to maximise profit by graphical method.

Example 2

A firm is engaged in animal breeding. The animals are to be given nutrition supplements everyday. There are two products A and B which contain the three required nutrients.

Nutrients	Quantity/unit		Minimum Requirement
	A	B	
1	72	12	216
2	6	24	72
3	40	20	200

Product cost per unit are: A: rs. 40; B: Rs. 80. Find out quantity of product A & B to be given to provide minimum nutritional requirement.

3. A firm produces two products P and Q. Daily production upper limit is 600 units for total production. But at least 300 total units must be produced every day. Machine hours consumption per unit is 6 for P and 2 for Q. At least 1200 machine hours must be used daily. Manufacturing costs per unit are Rs. 50 for P and Rs. 20 for Q. Find optimal solution for the LPP graphically.

4. A furniture manufacturer makes two products - tables and chairs. Processing of these products is done on two types of machines A and B. A chair requires 2 hours on machine type A and 6 hours on machine type B. A table requires 5 hours on machine type A and no time on Machine type B. There are 16 hours/day available on machine type A and 30 hours/day on machine type B. Profits gained by the manufacturer from a chair and a table are Rs. 2 and Rs. 10 respectively. What should be the daily production of each of the two products? Use graphical method of LPP to find the solution.